03 · Components
Parent: Board API [Container: Python / FastAPI]
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LEGEND / INTERFACE DATA
Blue boxes = components inside one Board API process.
Internal arrows = Python calls; ← = returned data.
Grey = external systems; green = persistent store.
VerifiedStudent = trusted {student_id, name, email}.
WorkshopSummary = {id,title,start_time,capacity,remaining,own_reservation}.
Attendee = {name,email}; only the workshop owner can request it.
Remaining = capacity – stored count (one consistent query snapshot).
Authentication failure → 401; wrong owner → 403.
R1–R6 refer to the customer rules on the student sheet.

Web UI
[Supporting container: HTML/CSS/JS]
Show authorised results and collect actions.

HTTPS/JSON + credential
Workshop actions; ← authorised DTOs / messages

Board API · Python / FastAPI application

Board Endpoints
[Component: FastAPI routes]
R1–R6 · Verify before routing every request.
Map typed results to HTTP responses.
Never trust client student/owner IDs.

create(verified, title, start_time, capacity)
publish / attendees(verified, workshop_id)
browse(verified) → WorkshopSummary[]
← Workshop / attendee list / forbidden

reserve(verified, workshop_id)
← Created(reservation,email) | Existing
| Rejected(reason)

verify(credential)
← VerifiedStudent{id,name,email} | Unverified
Run before Workshop Service / Reservation Rules

Workshop Service
[Component: Python service]
R2, R3, R5 · create / publish / browse / attendees
Generate unique ID; owner = verified student ID.
Create unpublished; title.strip() nonempty,
start_time required, integer capacity 1–20.
Only owner publishes or reads attendees.
Publishing creates no reservation.

Reservation Rules
[Component: Python service]
R4–R6 · reserve(identity, workshop_id)
Reject unverified; return repeat unchanged first.
Check published and capacity inside one lock.
Email only after committing a new seat.

Identity Gateway
[Component: Python HTTPS adapter]
R1, R4 · Verify every protected request.
Produce trusted VerifiedStudent or Unverified.
Unverified stops before domain calls.

save Workshop; publish if owner matches
read published summaries + own reservation
read attendees only after owner check
← Stored data / rejected

transaction(); tx.lockWorkshop(id)
tx.findExisting(id,student); tx.countSeats(id)
tx.insert(id,verified); tx.commit()
← Workshop / Reservation? / count / committed

New + committed only:
sendConfirmation(email, reservation, workshop)
← Sent | Unavailable

HTTPS/JSON: credential
← verified id/name/email or rejected

Board Repository
[Component: Python / psycopg]
R2–R5 · Persistent reads/writes
transaction() → ReservationTransaction
One connection, READ COMMITTED; lock workshop
before fresh reads, seat count and insert.
Every reservation writer uses this lock.

Mail Gateway
[Component: Python HTTPS adapter]
R6 · sendConfirmation(email, reservation, workshop)
Return Sent or Unavailable (error/timeout).
One attempt; never undo a saved seat.

Campus Identity
[External Software System]
Verify university identity.

PostgreSQL/TLS · SQL queries + transaction
SELECT workshop FOR UPDATE; fresh seat reads
INSERT / COMMIT; ← rows / acknowledgement

HTTPS/JSON: recipient email,
reservation ID + workshop title/time
← accepted / unavailable

Board Database
[Supporting container: PostgreSQL]
Persist Workshop + Reservation across restarts.
PRIMARY KEY Workshop.id and Reservation.id;
UNIQUE(workshop_id, student_id);
Reservation.workshop_id → Workshop.id foreign key.
CHECK capacity 1–20; nonempty title; NOT NULL fields.

University Mail
[External Software System]
Send confirmation email.